2.44 2nd Order DEs: Substitutions

Worked Solutions

Note for Integrating Factors: We can take the integrating factor to be any non-zero constant multiple of an integrating factor. For example, $e^{\int \frac{1}{x}\,dx} = e^{\ln|x|} = |x|$.

On any interval not containing $x = 0$, $|x|$ differs from x only by a constant factor (namely ± 1), so we may take the integrating factor to be x. So throughout these worked solutions, the modulus signs will be dropped when multiplying the differential equation through by the integrating factor.

1 a)

The 2nd order differential equation does not have constant coefficients, as $\frac{3}{x}$ is the coefficient of $\frac{dy}{dx}$.

1 b)

$$\begin{aligned} u = \frac{dy}{dx}, \quad \frac{du}{dx} &= \frac{d^2y}{dx^2} \\ \frac{d^2y}{dx^2} + \frac{3}{x}\frac{dy}{dx} &= x^2 \\ \Rightarrow \quad \frac{du}{dx} + \frac{3}{x}u &= x^2. \end{aligned}$$

1 c)

$$\begin{aligned} \frac{du}{dx} + \frac{3}{x}u &= x^2 \\ \text{Integrating Factor} &= e^{\int 3/x\,dx} = e^{3\ln|x|} = e^{\ln|x^3|} = |x^3| \\ \text{Let Integrating Factor} &= x^3 \\ x^3\frac{du}{dx} + \frac{3}{x}u \times x^3 &= x^2 \times x^3 \\ \Rightarrow x^3\frac{du}{dx} + 3x^2u &= x^5 \\ \Rightarrow \quad \frac{d}{dx}(x^3u) &= x^5 \\ \Rightarrow \quad x^3u &= \frac{1}{6}x^6 + c \\ \Rightarrow \quad u &= \frac{1}{6}x^3 + cx^{-3}. \end{aligned}$$

1 d)

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{6}x^3 + cx^{-3} \\ \Rightarrow \quad y &= \frac{1}{24}x^4 - \frac{1}{2}cx^{-2} + k \\ &= \frac{1}{24}x^4 + Ax^{-2} + k, \quad A = -\frac{1}{2}c. \end{aligned}$$

2 a)

The 2nd order differential equation does not have constant coefficients, as x is the coefficient of $\frac{d^2y}{dx^2}$.

2 b)

$$u = \frac{dy}{dx}, \quad \frac{du}{dx} = \frac{d^2y}{dx^2}$$
$$x\frac{d^2y}{dx^2} + \frac{1}{2}\frac{dy}{dx} = 3x^{3/2}$$
$$\Rightarrow \quad x\frac{du}{dx} + \frac{1}{2}u = 3x^{3/2}.$$

2 c)

$$x\frac{du}{dx} + \frac{1}{2}u = 3x^{3/2}$$
$$\frac{du}{dx} + \frac{1}{2x}u = 3x^{1/2}$$
$$\text{Integrating Factor} = e^{\int \frac{1}{2x}dx} = e^{\frac{1}{2}\ln|x|} = e^{\ln|x^{1/2}|} = |x^{1/2}|$$
$$\text{Let Integrating Factor} = x^{1/2}$$
$$x^{1/2}\frac{du}{dx} + \frac{1}{2x}u \times x^{1/2} = 3x^{1/2} \times x^{1/2}$$
$$\Rightarrow x^{1/2}\frac{du}{dx} + \frac{1}{2}x^{-1/2}u = 3x$$
$$\Rightarrow \quad \frac{d}{dx}(x^{1/2}u) = 3x$$
$$\Rightarrow \quad x^{1/2}u = \frac{3}{2}x^2 + c$$
$$\Rightarrow \quad u = \frac{3}{2}x^{3/2} + cx^{-1/2}.$$

2 d)

$$\frac{dy}{dx} = \frac{3}{2}x^{3/2} + cx^{-1/2}$$
$$\Rightarrow \quad y = \frac{3}{5}x^{5/2} + 2cx^{1/2} + k$$
$$= \frac{3}{5}x^{5/2} + Ax^{1/2} + k, \quad A = 2c.$$

3)

$$u = \frac{dy}{dx}, \quad \frac{du}{dx} = \frac{d^2y}{dx^2}$$
$$(2x+1)\frac{d^2y}{dx^2} + 3\frac{dy}{dx} = 4x + 2$$

$$
\begin{aligned}
\Rightarrow (2x+1)\frac{du}{dx} + 3u &= 4x+2 \\
(2x+1)\frac{du}{dx} + 3u &= 2(2x+1) \\
\frac{du}{dx} + \frac{3}{2x+1}u &= 2 \\
\text{Integrating Factor} &= e^{\int \frac{3}{2x+1}dx} \\
&= e^{\frac{3}{2}\ln|2x+1|} \\
&= e^{\ln|(2x+1)^{3/2}|} \\
&= |(2x+1)^{3/2}| \\
\text{Let Integrating Factor} &= (2x+1)^{3/2} \\
(2x+1)^{3/2}\frac{du}{dx} + 3(2x+1)^{1/2}u &= 2(2x+1)^{3/2} \\
\Rightarrow \frac{d}{dx}((2x+1)^{3/2}u) &= 2(2x+1)^{3/2} \\
(2x+1)^{3/2}u &= \frac{2}{5}(2x+1)^{5/2} + c \\
u &= \frac{2}{5}(2x+1) + c(2x+1)^{-3/2} \\
\frac{dy}{dx} &= \frac{2}{5}(2x+1) + c(2x+1)^{-3/2} \\
y &= \frac{1}{10}(2x+1)^2 - c(2x+1)^{-1/2} + k.
\end{aligned}
$$

4)

$$
\begin{aligned}
u = \frac{dy}{dx}, \quad \frac{du}{dx} &= \frac{d^2y}{dx^2} \\
(x^2+4)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} &= 4x \\
\Rightarrow (x^2+4)\frac{du}{dx} + 2xu &= 4x \\
\frac{du}{dx} + \frac{2x}{x^2+4}u &= \frac{4x}{x^2+4} \\
\text{Integrating Factor} &= e^{\int \frac{2x}{x^2+4}dx} \\
&= e^{\ln|x^2+4|} \\
&= e^{\ln|x^2+4|} \\
&= |x^2+4| \\
\text{Let Integrating Factor} &= x^2+4 \\
(x^2+4)\frac{du}{dx} + \frac{2x}{x^2+4}u(x^2+4) &= \frac{4x}{x^2+4}(x^2+4) \\
\Rightarrow (x^2+4)\frac{du}{dx} + 2xu &= 4x \\
\Rightarrow \frac{d}{dx}((x^2+4)u) &= 4x \\
(x^2+4)u &= 2x^2 + c \\
u &= \frac{2x^2+c}{x^2+4}
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{2(x^2+4)-8+c}{x^2+4} \\
&= 2 + \frac{c-8}{x^2+4} \\
\frac{dy}{dx} &= 2 + \frac{c-8}{4(1+\frac{1}{4}x^2)} \\
&= 2 + \frac{c-8}{4} \times \frac{1}{1+(\frac{1}{2}x)^2} \\
y &= 2x + \frac{c-8}{4} \times \frac{1}{1/2} \arctan(\tfrac{1}{2}x) + k \\
&= 2x + A\arctan(\tfrac{1}{2}x) + k, \quad A = \frac{c-8}{2} \\
\text{Substitute } y=1,\ x=0: &\quad 1 = 2\times 0 + A\arctan 0 + k \Rightarrow k = 1 \\
y &= 2x + A\arctan(\tfrac{1}{2}x) + 1 \\
\frac{dy}{dx} &= 2 + \frac{1}{2}A\frac{1}{1+(\frac{1}{2}x)^2} \\
\text{Substitute } \frac{dy}{dx}=0,\ x=0: &\quad 0 = 2 + \frac{1}{2}A\frac{1}{1+(\frac{1}{2}\times 0)^2} \Rightarrow A = -4 \\
\text{Particular Solution: } y &= 2x - 4\arctan(\tfrac{1}{2}x) + 1.
\end{aligned}
$$

5 a)

$$
\begin{aligned}
\frac{d}{dx}\left(\frac{x}{2}\tan x\right) &= \frac{x}{2}\sec^2 x + \frac{1}{2}\tan x \\
&= \frac{1}{2}x\sec^2 x + \frac{1}{2}\tan x.
\end{aligned}
$$

5 b)

$$
\begin{aligned}
u = \frac{dy}{dx}, \quad \frac{du}{dx} &= \frac{d^2y}{dx^2} \\
\cos x\frac{d^2y}{dx^2} - 2\sin x\frac{dy}{dx} &= \cos x \\
\Rightarrow \cos x\frac{du}{dx} - 2u\sin x &= \cos x \\
\frac{du}{dx} - 2\frac{\sin x}{\cos x}u &= 1 \\
\text{Integrating Factor} &= e^{\int -2\frac{\sin x}{\cos x}dx} \\
&= e^{2\ln|\cos x|} \\
&= e^{\ln|\cos^2 x|} \\
&= |\cos^2 x| \\
\text{Let Integrating Factor} &= \cos^2 x \\
\cos^2 x\frac{du}{dx} - 2\frac{\sin x}{\cos x}u\cos^2 x &= 1\times\cos^2 x \\
\Rightarrow \cos^2 x\frac{du}{dx} - 2\sin x\cos x u &= \cos^2 x
\end{aligned}
$$

$$\Rightarrow \frac{d}{dx}(u\cos^2 x) = \cos^2 x$$

$$\cos 2A \equiv 2\cos^2 A - 1 \Rightarrow \cos^2 A \equiv \frac{1}{2}\cos 2A + \frac{1}{2}$$

$$\therefore \quad \frac{d}{dx}(u\cos^2 x) = \frac{1}{2}\cos 2x + \frac{1}{2}$$

$$\Rightarrow u\cos^2 x = \frac{1}{4}\sin 2x + \frac{1}{2}x + c$$

$$\Rightarrow u = \frac{\frac{1}{4}\sin 2x + \frac{1}{2}x + c}{\cos^2 x}$$

$$\frac{dy}{dx} = \frac{\frac{1}{2}\sin x\cos x + \frac{1}{2}x + c}{\cos^2 x}$$

$$= \frac{\frac{1}{2}\sin x\cos x}{\cos^2 x} + \frac{\frac{1}{2}x}{\cos^2 x} + \frac{c}{\cos^2 x}$$

$$= \frac{1}{2}\tan x + \frac{1}{2}x\sec^2 x + c\sec^2 x$$

$$= \frac{d}{dx}(\frac{x}{2}\tan x) + c\sec^2 x$$

$$y = \frac{1}{2}x\tan x + c\tan x + k$$

Substitute $y = 2,\ x = 0:$ $\quad 2 = \frac{1}{2} \times 0 \times \tan 0 + c\tan 0 + k \Rightarrow k = 2$

$$y = \frac{1}{2}x\tan x + c\tan x + 2$$

$$\frac{dy}{dx} = \frac{1}{2}x\sec^2 x + \frac{1}{2}\tan x + c\sec^2 x$$

Substitute $\frac{dy}{dx} = 3,\ x = 0:$ $\quad 3 = \frac{1}{2} \times 0 \times \sec^2 0 + \frac{1}{2}\tan 0 + c\sec^2 0 \Rightarrow c = 3$

Particular Solution: $y = \frac{1}{2}x\tan x + 3\tan x + 2.$

6)

$$u = \frac{dy}{dx}, \quad \frac{du}{dx} = \frac{d^2y}{dx^2}$$

$$(x+2)(x+3)\frac{d^2y}{dx^2} + \frac{dy}{dx} = (x+3)^2$$

$$\Rightarrow (x+2)(x+3)\frac{du}{dx} + u = (x+3)^2$$

$$\frac{du}{dx} + P(x)u = Q(x)$$

$$(x+2)(x+3)\frac{du}{dx} + u = (x+3)^2$$

$$\Rightarrow \frac{du}{dx} + \frac{u}{(x+2)(x+3)} = \frac{x+3}{x+2}$$

Use Partial Fractions: $\quad \frac{1}{(x+2)(x+3)} \equiv \frac{A}{x+2} + \frac{B}{x+3}$

$$1 \equiv A(x+3) + B(x+2)$$

Let $x = -3:$ $\quad 1 = -B \Rightarrow B = -1$

Let $x = -2:$ $\quad 1 = A \Rightarrow A = 1$

$$\frac{du}{dx} + \left(\frac{1}{x+2} - \frac{1}{x+3}\right)u = \frac{x+3}{x+2}$$

$$\begin{aligned}
\text{Integrating Factor} &= e^{\int(\frac{1}{x+2}-\frac{1}{x+3})dx} \\
&= e^{\ln|x+2|-\ln|x+3|} \\
&= e^{\ln|\frac{x+2}{x+3}|} \\
&= |\frac{x+2}{x+3}| \\
\text{Let Integrating Factor} &= \frac{x+2}{x+3} \\
\frac{x+2}{x+3}\frac{du}{dx}+\frac{u}{(x+2)(x+3)}\frac{x+2}{x+3} &= \frac{x+3}{x+2}\frac{x+2}{x+3} \\
\Rightarrow \frac{x+2}{x+3}\frac{du}{dx}+\frac{u}{(x+3)^2} &= 1 \\
\Rightarrow \frac{d}{dx}(\frac{x+2}{x+3}u) &= 1 \\
\Rightarrow \frac{x+2}{x+3}u &= x+c \\
\Rightarrow u &= \frac{(x+3)(x+c)}{x+2} \\
\frac{dy}{dx} &= \frac{x^2+(3+c)x+3c}{x+2}
\end{aligned}$$

Or using polynomial division:

$$\begin{array}{r|l}
x+2 & x+1+c \\
 & x^2+(3+c)x+3c \\
 & x^2+2x \\
\hline
 & (1+c)x+3c \\
 & (1+c)x+2+2c \\
\hline
 & c-2
\end{array}$$

$$\begin{aligned}
\frac{x^2+(3+c)x+3c}{x+2} &\equiv \frac{x(x+2)+(x+2)+c(x+2)+c-2}{x+2} \\
&\equiv x+1+c+\frac{c-2}{x+2} \\
\frac{dy}{dx} &= x+1+c+\frac{c-2}{x+2} \\
y &= \frac{1}{2}x^2+(1+c)x+(c-2)\ln|x+2|+k \\
\text{Substitute } y=1,\ x=0: \quad 1 &= (c-2)\ln 2+k \\
\frac{dy}{dx} &= x+1+c+\frac{c-2}{x+2} \\
\text{Substitute } \frac{dy}{dx}=3,\ x=0: \quad 3 &= 0+1+c+\frac{c-2}{0+2} \\
2 &= c+\frac{c-2}{2} \\
4 &= 2c+c-2 \\
3c &= 6 \\
c &= 2 \\
1 &= (2-2)\ln 2+k \Rightarrow k=1
\end{aligned}$$

Particular Solution: $y = \frac{1}{2}x^2+3x+1.$

7 a)

$$\frac{d}{dx}\left(x^4\frac{dy}{dx}\right) = x^4\frac{d^2y}{dx^2} + 4x^3\frac{dy}{dx}.$$

7 b)

$$\begin{aligned}
u &= x^4\frac{dy}{dx} \\
\Rightarrow \frac{du}{dx} &= x^4\frac{d^2y}{dx^2} + 4x^3\frac{dy}{dx} \\
\text{So: } x^4(x^2+1)\frac{d^2y}{dx^2} + 4x^3(x^2+1)\frac{dy}{dx} - 2x^5\frac{dy}{dx} &= 2x(x^2+1)^2 \\
\Rightarrow (x^2+1)\frac{du}{dx} - 2x^5\frac{dy}{dx} &= 2x(x^2+1)^2 \\
\text{As } u = x^4\frac{dy}{dx} \Rightarrow \frac{dy}{dx} &= \frac{u}{x^4} \\
\Rightarrow (x^2+1)\frac{du}{dx} - 2x^5 \times \frac{u}{x^4} &= 2x(x^2+1)^2 \\
\Rightarrow (x^2+1)\frac{du}{dx} - 2xu &= 2x(x^2+1)^2 \\
\frac{du}{dx} - \frac{2x}{x^2+1}u &= 2x(x^2+1) \\
\text{Integrating Factor} &= e^{\int -\frac{2x}{x^2+1}dx} \\
&= e^{-\ln|x^2+1|} \\
&= e^{\ln|(x^2+1)^{-1}|} \\
&= \left|\frac{1}{x^2+1}\right| \\
\text{Let Integrating Factor} &= \frac{1}{x^2+1} \\
\frac{1}{x^2+1}\frac{du}{dx} - \frac{2x}{x^2+1}u \times \frac{1}{x^2+1} &= 2x(x^2+1) \times \frac{1}{x^2+1} \\
\Rightarrow \frac{1}{x^2+1}\frac{du}{dx} - \frac{2x}{(x^2+1)^2}u &= 2x \\
\Rightarrow \frac{d}{dx}\left(\frac{1}{x^2+1}u\right) &= 2x \\
\Rightarrow \frac{1}{x^2+1}u &= x^2 + c \\
\Rightarrow u &= (x^2+1)(x^2+c) \\
x^4\frac{dy}{dx} &= x^4 + (1+c)x^2 + c \\
\Rightarrow \frac{dy}{dx} &= 1 + \frac{1+c}{x^2} + \frac{c}{x^4} \\
\Rightarrow y &= x - \frac{1+c}{x} - \frac{c}{3x^3} + k.
\end{aligned}$$

8 a)

$$u = \frac{dy}{dx} - 3x$$

$$\Rightarrow \frac{du}{dx} = \frac{d^2y}{dx^2} - 3$$

$$\text{So } \frac{dy}{dx} = u + 3x \text{ and } \frac{d^2y}{dx^2} = \frac{du}{dx} + 3$$

$$(1+x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} = 3 - 3x^2$$

$$\Rightarrow (1+x^2)(\frac{du}{dx} + 3) - 2x(u+3x) = 3 - 3x^2$$

$$\Rightarrow (1+x^2)\frac{du}{dx} + 3(1+x^2) - 2xu - 6x^2 = 3 - 3x^2$$

$$\Rightarrow (1+x^2)\frac{du}{dx} + 3 + 3x^2 - 2xu - 6x^2 = 3 - 3x^2$$

$$\Rightarrow (1+x^2)\frac{du}{dx} - 2xu = 0$$

$$\Rightarrow (1+x^2)\frac{du}{dx} = 2xu.$$

8 b)

$$\text{Use Separation of Variables: } \int \frac{1}{u}\,du = \int \frac{2x}{1+x^2}\,dx$$

$$\Rightarrow \ln|u| = \ln|1+x^2| + c$$

$$\Rightarrow u = \pm e^{\ln|1+x^2|+c}$$

$$\Rightarrow u = A(1+x^2)$$

$$\text{Now } u = \frac{dy}{dx} - 3x: \quad \frac{dy}{dx} - 3x = A + Ax^2$$

$$\text{Substitute } \frac{dy}{dx} = 2,\ x = 0: \quad 2 - 3\times 0 = A + A\times 0^2 \Rightarrow A = 2$$

$$\therefore \quad \frac{dy}{dx} - 3x = 2 + 2x^2$$

$$\Rightarrow \frac{dy}{dx} = 2x^2 + 3x + 2$$

$$\Rightarrow y = \frac{2}{3}x^3 + \frac{3}{2}x^2 + 2x + k$$

$$\text{Substitute } y = 1,\ x = 0: \quad 1 = \frac{2}{3}\times 0^3 + \frac{3}{2}\times 0^2 + 2\times 0 + k \Rightarrow k = 1$$

$$\therefore y = \frac{2}{3}x^3 + \frac{3}{2}x^2 + 2x + 1.$$

9 a)

$$u = \frac{dy}{dx} + 3y$$

$$\Rightarrow \frac{du}{dx} = \frac{d^2y}{dx^2} + 3\frac{dy}{dx}$$

$$\text{So } \frac{dy}{dx} = u - 3y \text{ and } \frac{d^2y}{dx^2} = \frac{du}{dx} - 3\frac{dy}{dx}$$

$$= \frac{du}{dx} - 3(u - 3y) = \frac{du}{dx} - 3u + 9y$$

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = e^{-3x}$$
$$\Rightarrow (\frac{du}{dx} - 3u + 9y) + 6(u - 3y) + 9y = e^{-3x}$$
$$\Rightarrow \frac{du}{dx} - 3u + 9y + 6u - 18y + 9y = e^{-3x}$$
$$\Rightarrow \frac{du}{dx} + 3u = e^{-3x}.$$

9 b)

$$\frac{du}{dx} + 3u = e^{-3x}$$
$$\text{Integrating Factor} = e^{\int 3dx}$$
$$= e^{3x}$$
$$e^{3x}\frac{du}{dx} + 3e^{3x}u = e^{-3x} \times e^{3x}$$
$$\Rightarrow e^{3x}\frac{du}{dx} + 3e^{3x}u = 1$$
$$\Rightarrow \frac{d}{dx}(e^{3x}u) = 1$$
$$\Rightarrow e^{3x}u = x + c$$
$$\Rightarrow u = (x + c)e^{-3x}.$$

9 c)

$$\text{Now } u = \frac{dy}{dx} + 3y \text{ so:}$$
$$\frac{dy}{dx} + 3y = (x + c)e^{-3x}$$
$$\text{Integrating Factor} = e^{\int 3dx}$$
$$= e^{3x}$$
$$e^{3x}\frac{dy}{dx} + 3e^{3x}y = (x + c)e^{-3x} \times e^{3x}$$
$$\Rightarrow e^{3x}\frac{dy}{dx} + 3e^{3x}y = x + c$$
$$\Rightarrow \frac{d}{dx}(e^{3x}y) = x + c$$
$$\Rightarrow e^{3x}y = \frac{1}{2}x^2 + cx + k$$
$$\Rightarrow y = (\frac{1}{2}x^2 + cx + k)e^{-3x}.$$

10 a)

$$u = \ln x$$

$$\Rightarrow \frac{du}{dx} = \frac{1}{x}$$

By the Chain Rule, we can write: $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{1}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{x}\frac{dy}{du}.$$

10 b)

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(\frac{1}{x}\frac{dy}{du})$$

Use the Product Rule and note that: $\frac{d}{dx}(\frac{dy}{du}) = \frac{d}{dx}(y'(u)) = \frac{du}{dx}y''(u) = \frac{d^2y}{du^2}\frac{du}{dx}$

Then: $\frac{d^2y}{dx^2} = \frac{d}{dx}(\frac{1}{x}\frac{dy}{du})$

$$= -\frac{1}{x^2}\frac{dy}{du} + \frac{1}{x}\frac{d^2y}{du^2}\frac{du}{dx}$$

Recall that $\frac{du}{dx} = \frac{1}{x}$

$$\frac{d^2y}{dx^2} = -\frac{1}{x^2}\frac{dy}{du} + \frac{1}{x}\frac{d^2y}{du^2} \times \frac{1}{x}$$

$$= -\frac{1}{x^2}\frac{dy}{du} + \frac{1}{x^2}\frac{d^2y}{du^2}$$

$$= \frac{1}{x^2}(\frac{d^2y}{du^2} - \frac{dy}{du}) \text{ as required.}$$

11)

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 4y = 0$$

$$\Rightarrow x^2 \times \frac{1}{x^2}(\frac{d^2y}{du^2} - \frac{dy}{du}) + x \times \frac{1}{x}\frac{dy}{du} - 4y = 0$$

$$\Rightarrow \frac{d^2y}{du^2} - \frac{dy}{du} + \frac{dy}{du} - 4y = 0$$

$$\Rightarrow \frac{d^2y}{du^2} - 4y = 0$$

Auxiliary Equation: $m^2 - 4 = 0$

$$\Rightarrow (m-2)(m+2) = 0$$

$$\Rightarrow m = 2 \text{ or } m = -2$$

$$\therefore y = Ae^{2u} + Be^{-2u}$$

Now $x = e^u$ so: $y = A(e^u)^2 + B(e^u)^{-2}$

$$\Rightarrow y = Ax^2 + Bx^{-2}.$$

12)

$$x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 8y = 4$$
$$\Rightarrow x^2 \times \frac{1}{x^2}(\frac{d^2y}{du^2} - \frac{dy}{du}) - x \times \frac{1}{x}\frac{dy}{du} - 8y = 4$$
$$\Rightarrow \frac{d^2y}{du^2} - \frac{dy}{du} - \frac{dy}{du} - 8y = 4$$
$$\Rightarrow \frac{d^2y}{du^2} - 2\frac{dy}{du} - 8y = 4$$
$$\text{Homogeneous Equation: } \frac{d^2y}{du^2} - 2\frac{dy}{du} - 8y = 0$$
$$\text{Auxiliary Equation: } m^2 - 2m - 8 = 0$$
$$\Rightarrow (m-4)(m+2) = 0$$
$$\Rightarrow m = 4 \text{ or } m = -2$$
$$\text{Complementary Function: } y = Ae^{4u} + Be^{-2u}$$
$$\text{Try this Particular Integral: } y = k, \quad \frac{dy}{du} = 0, \quad \frac{d^2y}{du^2} = 0$$
$$\frac{d^2y}{du^2} - 2\frac{dy}{du} - 8y = 4$$
$$\Rightarrow 0 - 2(0) - 8k = 4$$
$$\Rightarrow -8k = 4$$
$$\Rightarrow k = -\frac{1}{2}$$
$$\text{Particular Integral: } y = -\frac{1}{2}$$
$$\text{General Solution: } y = Ae^{4u} + Be^{-2u} - \frac{1}{2}$$
$$\text{Now } x = e^u \text{ so: } y = A(e^u)^4 + B(e^u)^{-2} - \frac{1}{2}$$
$$\Rightarrow y = Ax^4 + Bx^{-2} - \frac{1}{2}.$$

13)

$$x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} - 3y = 5x^2$$
$$\Rightarrow x^2 \times \frac{1}{x^2}(\frac{d^2y}{du^2} - \frac{dy}{du}) + 3x \times \frac{1}{x}\frac{dy}{du} - 3y = 5(e^u)^2$$
$$\Rightarrow \frac{d^2y}{du^2} - \frac{dy}{du} + 3\frac{dy}{du} - 3y = 5e^{2u}$$
$$\Rightarrow \frac{d^2y}{du^2} + 2\frac{dy}{du} - 3y = 5e^{2u}$$
$$\text{Homogeneous Equation: } \frac{d^2y}{du^2} + 2\frac{dy}{du} - 3y = 0$$
$$\text{Auxiliary Equation: } m^2 + 2m - 3 = 0$$
$$\Rightarrow (m+3)(m-1) = 0$$
$$\Rightarrow m = -3 \text{ or } m = 1$$

$$\begin{aligned}
\text{Complementary Function: } & y = Ae^{-3u} + Be^{u} \\
\text{Try this Particular Integral: } & y = ke^{2u} \\
& \frac{dy}{du} = 2ke^{2u}, \quad \frac{d^2y}{du^2} = 4ke^{2u} \\
& \frac{d^2y}{du^2} + 2\frac{dy}{du} - 3y = 5e^{2u} \\
\Rightarrow & (4ke^{2u}) + 2(2ke^{2u}) - 3(ke^{2u}) = 5e^{2u} \\
\Rightarrow & (4 + 4 - 3)ke^{2u} = 5e^{2u} \\
\Rightarrow & 5ke^{2u} = 5e^{2u} \\
\Rightarrow & k = 1 \\
\text{Particular Integral: } & y = e^{2u} \\
\text{General Solution: } & y = Ae^{-3u} + Be^{u} + e^{2u} \\
\text{Now } x = e^u \text{ so: } & y = A(e^u)^{-3} + Be^{u} + (e^u)^2 \\
\Rightarrow & y = Ax^{-3} + Bx + x^2 \\
\text{Substitute } y = 2,\ x = 1: \quad & 2 = A \times 1^{-3} + B \times 1 + 1^2 \\
& A + B = 1 \qquad (1) \\
& \frac{dy}{dx} = -3Ax^{-4} + B + 2x \\
\text{Substitute } \frac{dy}{dx} = 1,\ x = 1: \quad & 1 = -3A \times 1^{-4} + B + 2 \times 1 \\
& -3A + B = -1 \qquad (2) \\
\text{Solving (1) and (2): } & A = \frac{1}{2}, \quad B = \frac{1}{2} \\
\text{Particular Solution: } & y = \frac{1}{2x^3} + \frac{1}{2}x + x^2.
\end{aligned}$$

14 a)

$$\begin{aligned}
& x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = 12 + 10\cos(\ln x) \\
\Rightarrow & x^2 \times \frac{1}{x^2}(\frac{d^2y}{dt^2} - \frac{dy}{dt}) - 4x \times \frac{1}{x}\frac{dy}{dt} + 6y = 12 + 10\cos(\ln e^t) \\
\Rightarrow & \frac{d^2y}{dt^2} - \frac{dy}{dt} - 4\frac{dy}{dt} + 6y = 12 + 10\cos t \\
\Rightarrow & \frac{d^2y}{dt^2} - 5\frac{dy}{dt} + 6y = 12 + 10\cos t.
\end{aligned}$$

14 b)

$$\begin{aligned}
\text{Homogeneous Equation: } & \frac{d^2y}{dt^2} - 5\frac{dy}{dt} + 6y = 0 \\
\text{Auxiliary Equation: } & m^2 - 5m + 6 = 0 \\
\Rightarrow & (m-2)(m-3) = 0 \\
\Rightarrow & m = 2 \text{ or } m = 3 \\
\text{Complementary Function: } & y = Ae^{2t} + Be^{3t}
\end{aligned}$$

$$\text{Try this Particular Integral: } y = k + p\cos t + q\sin t$$
$$\frac{dy}{dt} = -p\sin t + q\cos t$$
$$\frac{d^2y}{dt^2} = -p\cos t - q\sin t$$
$$\frac{d^2y}{dt^2} - 5\frac{dy}{dt} + 6y = 12 + 10\cos t$$

$$\Rightarrow (-p\cos t - q\sin t) - 5(-p\sin t + q\cos t) + 6(k + p\cos t + q\sin t) = 12 + 10\cos t$$

$$\Rightarrow 6k + (-p - 5q + 6p)\cos t + (-q + 5p + 6q)\sin t = 12 + 10\cos t$$
$$\Rightarrow 6k + (5p - 5q)\cos t + (5p + 5q)\sin t = 12 + 10\cos t$$
$$\text{Compare Coefficients: } 6k = 12 \Rightarrow k = 2$$
$$5p - 5q = 10 \quad (1)$$
$$5p + 5q = 0 \quad (2)$$
$$\text{Solve (1) and (2): } p = 1, \quad q = -1$$
$$\text{Particular Integral: } y = 2 + \cos t - \sin t$$
$$\text{General Solution: } y = Ae^{2t} + Be^{3t} + 2 + \cos t - \sin t$$
$$\text{Now } x = e^t \Rightarrow t = \ln x \text{ so: } y = Ae^{2\ln x} + Be^{3\ln x} + 2 + \cos(\ln x) - \sin(\ln x)$$
$$\Rightarrow y = Ae^{\ln x^2} + Be^{\ln x^3} + 2 + \cos(\ln x) - \sin(\ln x)$$
$$\Rightarrow y = Ax^2 + Bx^3 + 2 + \cos(\ln x) - \sin(\ln x).$$

15 a)

$$x = e^{3t}$$
$$\Rightarrow \frac{dx}{dt} = 3e^{3t}$$
$$\Rightarrow \frac{dt}{dx} = \frac{1}{3e^{3t}} = \frac{1}{3x}$$
$$\text{By the Chain Rule, we can write: } \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$
$$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \times \frac{1}{3x}$$
$$\Rightarrow \frac{dy}{dx} = \frac{1}{3x}\frac{dy}{dt}.$$

15 b)

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{3x}\frac{dy}{dt}\right)$$
$$\text{Use the Product Rule and note that: } \frac{d}{dx}\left(\frac{dy}{dt}\right) = \frac{d}{dx}(y'(t)) = \frac{dt}{dx}y''(t) = \frac{d^2y}{dt^2}\frac{dt}{dx}$$
$$\text{Then: } \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{3x}\frac{dy}{dt}\right)$$
$$= -\frac{1}{3x^2}\frac{dy}{dt} + \frac{1}{3x}\frac{d^2y}{dt^2}\frac{dt}{dx}$$

$$\text{Recall that } \frac{dt}{dx} = \frac{1}{3x}$$
$$\frac{d^2y}{dx^2} = -\frac{1}{3x^2}\frac{dy}{dt} + \frac{1}{3x}\frac{d^2y}{dt^2} \times \frac{1}{3x}$$
$$= -\frac{1}{3x^2}\frac{dy}{dt} + \frac{1}{9x^2}\frac{d^2y}{dt^2}$$
$$= \frac{1}{9x^2}(\frac{d^2y}{dt^2} - 3\frac{dy}{dt}) \text{ as required.}$$

16 a)

$$9x^3\frac{d^2y}{dx^2} + 2xy = x(\ln x)^2 + 20$$
$$\Rightarrow 9x^3 \times \frac{1}{9x^2}(\frac{d^2y}{dt^2} - 3\frac{dy}{dt}) + 2xy = x(\ln x)^2 + 20$$
$$\Rightarrow x(\frac{d^2y}{dt^2} - 3\frac{dy}{dt}) + 2xy = x(\ln x)^2 + 20$$
$$\Rightarrow e^{3t}(\frac{d^2y}{dt^2} - 3\frac{dy}{dt}) + 2e^{3t}y = e^{3t}(\ln(e^{3t}))^2 + 20$$
$$\Rightarrow e^{3t}\frac{d^2y}{dt^2} - 3e^{3t}\frac{dy}{dt} + 2e^{3t}y = e^{3t}(3t)^2 + 20$$
$$\Rightarrow \frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = 9t^2 + 20e^{-3t}.$$

16 b)

$$\text{Homogeneous Equation: } \frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = 0$$
$$\text{Auxiliary Equation: } m^2 - 3m + 2 = 0$$
$$\Rightarrow (m-1)(m-2) = 0$$
$$\Rightarrow m = 1 \text{ or } m = 2$$
$$\text{Complementary Function: } y = Ae^t + Be^{2t}$$
$$\text{Try this Particular Integral: } y = pt^2 + qt + r + ke^{-3t}$$
$$\frac{dy}{dt} = 2pt + q - 3ke^{-3t}$$
$$\frac{d^2y}{dt^2} = 2p + 9ke^{-3t}$$
$$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = 9t^2 + 20e^{-3t}$$
$$\Rightarrow (2p + 9ke^{-3t}) - 3(2pt + q - 3ke^{-3t}) + 2(pt^2 + qt + r + ke^{-3t}) = 9t^2 + 20e^{-3t}$$
$$\Rightarrow (2p)t^2 + (-6p + 2q)t + (2p - 3q + 2r) + (9 + 9 + 2)ke^{-3t} = 9t^2 + 20e^{-3t}$$
$$\Rightarrow (2p)t^2 + (-6p + 2q)t + (2p - 3q + 2r) + 20ke^{-3t} = 9t^2 + 20e^{-3t}$$
$$\text{Compare Coefficients: } 2p = 9 \Rightarrow p = \frac{9}{2}$$
$$-6 \times \frac{9}{2} + 2q = 0 \Rightarrow q = \frac{27}{2}$$
$$2 \times \frac{9}{2} - 3 \times \frac{27}{2} + 2r = 0 \Rightarrow r = \frac{63}{4}$$

$$20k = 20 \Rightarrow k = 1$$

Particular Integral: $y = \frac{9}{2}t^2 + \frac{27}{2}t + \frac{63}{4} + e^{-3t}$

$$\text{General Solution: } y = Ae^t + Be^{2t} + \frac{9}{2}t^2 + \frac{27}{2}t + \frac{63}{4} + e^{-3t}$$

Now $x = e^{3t} \Rightarrow t = \frac{1}{3}\ln x$ so:

$$y = Ae^{\frac{1}{3}\ln x} + Be^{\frac{2}{3}\ln x} + \frac{9}{2}(\frac{1}{3}\ln x)^2 + \frac{27}{2}(\frac{1}{3}\ln x) + \frac{63}{4} + \frac{1}{x}$$

$$\Rightarrow y = Ae^{\ln x^{1/3}} + Be^{\ln x^{2/3}} + \frac{9}{2}(\frac{1}{3}\ln x)^2 + \frac{27}{2}(\frac{1}{3}\ln x) + \frac{63}{4} + \frac{1}{x}$$

$$\Rightarrow y = Ax^{1/3} + Bx^{2/3} + \frac{1}{2}(\ln x)^2 + \frac{9}{2}\ln x + \frac{63}{4} + \frac{1}{x}.$$

17 a)

$$\begin{aligned} y &= u\cos x \\ \Rightarrow \frac{dy}{dx} &= -u\sin x + \cos x\frac{du}{dx} \\ \Rightarrow \frac{d^2y}{dx^2} &= -u\cos x - \sin x\frac{du}{dx} + \cos x\frac{d^2u}{dx^2} - \sin x\frac{du}{dx} \\ &= -u\cos x - 2\sin x\frac{du}{dx} + \cos x\frac{d^2u}{dx^2} \end{aligned}$$

So: $\cos^2 x\frac{d^2y}{dx^2} + \sin 2x\frac{dy}{dx} + 2y = \sin 2x\cos^3 x$

$$\Rightarrow \cos^2 x(-u\cos x - 2\sin x\frac{du}{dx} + \cos x\frac{d^2u}{dx^2}) + 2\sin x\cos x(-u\sin x + \cos x\frac{du}{dx}) + 2(u\cos x) = 2\sin x\cos x \times \cos^3 x$$

$$\Rightarrow -u\cos^3 x - 2\sin x\cos^2 x\frac{du}{dx} + \cos^3 x\frac{d^2u}{dx^2} - 2u\sin^2 x\cos x + 2\sin x\cos^2 x\frac{du}{dx} + 2u\cos x = 2\sin x\cos^4 x$$

$$\begin{aligned} \Rightarrow -u\cos^2 x + \cos^2 x\frac{d^2u}{dx^2} - 2u\sin^2 x + 2u &= 2\sin x\cos^3 x \\ \Rightarrow \cos^2 x\frac{d^2u}{dx^2} + (2 - \cos^2 x - 2\sin^2 x)u &= 2\sin x\cos^3 x \end{aligned}$$

$$\Rightarrow \cos^2 x\frac{d^2u}{dx^2} + (2\sin^2 x + 2\cos^2 x - \cos^2 x - 2\sin^2 x)u = 2\sin x\cos^3 x$$
$$\Rightarrow \cos^2 x\frac{d^2u}{dx^2} + \cos^2 xu = 2\sin x\cos^3 x$$
$$\Rightarrow \frac{d^2u}{dx^2} + u = 2\sin x\cos x$$
$$\Rightarrow \frac{d^2u}{dx^2} + u = \sin 2x.$$

17 b)

$$\text{Homogeneous Equation: } \frac{d^2u}{dx^2} + u = 0$$
$$\text{Auxiliary Equation: } m^2 + 1 = 0$$
$$\Rightarrow m^2 = -1$$
$$\Rightarrow m = \pm i$$
$$\text{Complementary Function: } u = A\cos x + B\sin x$$
$$\text{Try this Particular Integral: } u = p\cos 2x + q\sin 2x$$
$$\frac{du}{dx} = -2p\sin 2x + 2q\cos 2x$$
$$\frac{d^2u}{dx^2} = -4p\cos 2x - 4q\sin 2x$$
$$\frac{d^2u}{dx^2} + u = \sin 2x$$
$$\Rightarrow (-4p\cos 2x - 4q\sin 2x) + (p\cos 2x + q\sin 2x) = \sin 2x$$
$$\Rightarrow (-4p + p)\cos 2x + (-4q + q)\sin 2x = \sin 2x$$
$$\Rightarrow (-3p)\cos 2x + (-3q)\sin 2x = \sin 2x$$
$$\text{Compare Coefficients: } -3p = 0 \Rightarrow p = 0$$
$$-3q = 1 \Rightarrow q = -\frac{1}{3}$$
$$\text{Particular Integral: } u = -\frac{1}{3}\sin 2x$$
$$\text{General Solution: } u = A\cos x + B\sin x - \frac{1}{3}\sin 2x$$
$$\text{Now } y = u\cos x \Rightarrow u = \frac{y}{\cos x} \text{ so:}$$
$$\frac{y}{\cos x} = A\cos x + B\sin x - \frac{1}{3}\sin 2x$$
$$\Rightarrow y = A\cos^2 x + B\sin x\cos x - \frac{1}{3}\sin 2x\cos x.$$